

Linear Algebra: The Gram-Schmidt Process

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Outline

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- 2 Vector Spaces
- 3 Basis
- 4 Gram-Schmidt Process
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Linear Independence

An indexed set of vectors $\{v_1, v_2, \dots, v_p\}$ in a vector space V is said to be **linearly independent** if the vector equation

$$c_1 v_1 + c_2 v_2 + \cdots + c_p v_p = 0$$

has only the trivial solution:

$$c_1 = c_2 = \cdots = c_p = 0.$$

For example, the standard basis of $\mathbb{R}^n$ is linearly independent.

Orthogonality

Two vectors u and v in $\mathbb{R}^n$ are **orthogonal** if their dot product is zero:

$$u \cdot v = 0.$$

This concept is fundamental, particularly in processes like Gram–Schmidt where an orthogonal basis is constructed.

Vector Spaces (Part I)

A **vector space** V over the real numbers $\mathbb{R}$ is a non-empty set of objects (called **vectors**) on which two operations are defined: vector addition and scalar multiplication. These operations must satisfy the following rules for all vectors $u, v, w \in V$ and scalars $c, d \in \mathbb{R}$:

- 1 **Closure under addition:** $u + v \in V$.
- 2 **Commutativity:** $u + v = v + u$.
- 3 **Associativity:** $(u + v) + w = u + (v + w)$.
- 4 **Existence of the zero vector:** There exists a $0 \in V$ such that $u + 0 = u$.
- 5 **Existence of additive inverses:** For each $u \in V$, there exists $-u \in V$ with $u + (-u) = 0$.

Vector Spaces (Part II)

In addition, V must satisfy:

- 6 **Closure under scalar multiplication:** $cu \in V$.
- 7 **Distributivity over vector addition:** $c(u + v) = cu + cv$.
- 8 **Distributivity over scalar addition:** $(c + d)u = cu + du$.
- 9 **Associativity of scalar multiplication:** $c(du) = (cd)u$.
- 10 **Identity element of scalar multiplication:** $1u = u$.

The dimension of a vector space is determined by the number of components in a singular vector ele.

Subspace via Span (Thm. 1)

Theorem 1:

If $v_1, \dots, v_p$ are vectors in a vector space V , then

$$\text{Span}\{v_1, \dots, v_p\}$$

is a subspace of V .

Column Spaces (Thm. 2)

The **column space** of an $m \times n$ matrix A , denoted as $\text{Col } A$, is the set of all linear combinations of the columns of A . If

$$A = [a_1, a_2, \dots, a_n],$$

then

$$\text{Col } A = \text{Span}\{a_1, a_2, \dots, a_n\}.$$

Theorem 2:

The column space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^m$.

Null Spaces (Thm. 3)

The **null space** of an $m \times n$ matrix A , denoted $\text{Nul } A$, is the set of all solutions to the homogeneous equation

$$A\mathbf{x} = 0.$$

In set notation,

$$\text{Nul } A = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = 0\}.$$

Theorem 3:

The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.

Spanning Sets

Definition of spanning set.

Example

Examples of spanning sets.

Basis

Definition of Basis.

Example

Examples of Bases.

Orthogonal Basis

Definition

An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis that is also an orthogonal set, i.e., every pair of distinct vectors in the basis is orthogonal.

Orthogonal bases simplify calculations and provide insight into the structure of the subspace.

Orthogonal Projection

Definition and formula for orthogonal projection.

Orthogonal Projection

Further details on orthogonal projection.

Gram-Schmidt Process

Theorem

Given a basis $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p\}$ for a subspace W of $\mathbb{R}^n$, define

$$\mathbf{v}_1 = \mathbf{x}_1,$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1,$$

$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2,$$

$$\vdots$$

$$\mathbf{v}_p = \mathbf{x}_p - \sum_{j=1}^{p-1} \frac{\mathbf{x}_p \cdot \mathbf{v}_j}{\mathbf{v}_j \cdot \mathbf{v}_j} \mathbf{v}_j.$$

Gram-Schmidt Process

Theorem

Then $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is an orthogonal basis for W . Moreover, for each $1 \leq k \leq p$, we have

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{x}_k\} = \text{Span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}.$$

Gram-Schmidt Process Example

Example

Let $W = \text{Span}\{\mathbf{x}_1, \mathbf{x}_2\}$, where

$$\mathbf{x}_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Apply the Gram-Schmidt process to obtain an orthogonal basis for W .

Thank You!

Questions?